

which is assumed to be an independent Poisson process. This establishes the relationship between the microscopic dynamics of the limit order book and the long-term dynamics of the asset price process. Unlike traditional methods that model price dynamics using one-dimensional stochastic processes, the derived model is a two-dimensional stochastic process where the additional dimension represents the latest price change. The parameters of the proposed model can be estimated directly from the order arrival and cancellation rate describing the incoming flow of orders. Additionally, the model also allows us to efficiently compute several quantities of interest of the asset price dynamics (i.e. volatility, return distribution, and first-passage probability) without resorting to simulation.

The proposed parameter estimation method tends to have increased error when the arrival rate of limit orders is small when compared to the arrival rate of market orders as well as when the order cancellation rate is getting smaller. The merit of the proposed framework is the ability to accurately predict the long-term behaviour of the limit order market when we get the right parameters. Hence future work will be focused on developing a better method for estimating the distribution of the number of the orders at the reference price which is the cause of this error. Additionally, we also look forward to investigating the validity of the proposed model by estimating model parameters from a real data set and comparing the prediction with the empirical results.

Chapter 7

Order placement strategy

To illustrate a way to utilise the models of execution probability studied in this dissertation to make order placement decisions, this chapter proposes a new framework for making order placement decisions based on the trade-off between the profit gained from better execution prices and the risk of non-execution that uses the developed execution probability model to balance this trade-off. The result obtained from applying the proposed framework to make order placement for liquidity traders who need to transact their order before the end of a deadline in the historical dataset obtained from the Multi Commodity of India and the New York Stock Exchange indicates that the proposed framework has better performance than the best static order placement strategy for all instruments in the Multi Commodity of India, while it beat the best static strategy only in two out of six cases studied in the New York Stock Exchange. Although the proposed framework cannot beat the best static strategy in all cases, the improvement gained from the proposed framework when it can beat the best static strategy is very significant.

7.1 Introduction

To illustrate the application of the execution probability model developed in this study, this chapter presents and investigates an order placement strategy that uses the developed model to make order placement decisions. This decision is very important especially for traders who trade in limit-order markets, where traders can freely specify the price at which they want to trade. On one hand, traders would prefer to place their orders far away from the current best price as this will increase their payoff. On the other hand, the farther away from the best price, the lower the chance that the order will be executed. Consequently, traders have to find the right tradeoff between these two opposite choices in order to maximise the profit gained from the trade.

In reality, an order submission strategy that a trader selects normally depends on the trading problem he is trying to solve. As suggested by Harris [42], three main trading problems frequently faced by traders are: (i) *the liquidity trader problem* considers how a liquidity trader who must fill his order before some deadlines should trade, (ii) *the informed trader problem* considers how an informed trader who receives a single signal about asset value should trade before his information becomes obsolete and (iii) *the value-*

motivated trader problem considers how a trader who continuously estimates security value should trade. Specifically, liquidity traders must fill their order before some deadline which may arise when they need to invest or disinvest their cash flow. The main objective of these traders is to obtain the best price for their trades by carefully choosing their order submission strategies. On the other hand, informed traders, who have private information about the underlying value of the asset, want to profitably trade on their information. Although informed traders do have a trading deadline, which is the time their information becomes obsolete, they did not have to fill their orders before the deadline like liquidity traders. In fact, informed traders will trade only when it is profitable to do so. Like informed traders, value-motivated traders also have private information about the value of assets. However, unlike informed traders, they do not have a specific deadline and are assumed to trade repeatedly in the market since they receive continuous information about the values.

This chapter presents a framework to solve these decision problems based on the developed execution probability model. Section 7.2 starts the chapter by giving a brief review on previously proposed methods to solve this problem. In Section 7.3, the proposed framework that utilised the developed execution probability model to make order placement decision is presented. Section 7.3 then gives a detailed discussion on how to utilise the proposed framework to make order placement decisions for liquidity traders, as well as analysing the result obtained from applying the proposed framework to make order placement decisions in a historical dataset. Finally, the conclusion of the results investigated in this section are summarised in Section 7.5.

7.2 Previous work

Order placement strategies previously proposed in the literature can be classified into two main categories: (i) static order placement strategies and (ii) dynamic order placement strategies. Static order submission strategies view this problem as a one-shot game where traders can make their order decision only once. If they decide to submit a limit order, no additional change can be made to the order and it will stay in the order book until it is executed or the end of the trading period is reached. Conversely, dynamic order submission strategies allow traders to cancel or make changes to their orders before the order expires or is executed [42, 87]. Empirically, traders change their order submission as market conditions change. They continuously monitor the market and make appropriate changes to their orders whenever necessary. For example, to reduce the execution risk, they may convert their limit orders to market orders when the demand for immediacy increases. They may also reprice or cancel their limit orders when the underlying value of the asset changes to manage the adverse selection cost. Hence, it is more appropriate to model this decision with dynamic strategies than with static strategies.

This section briefly describes related work in order submission strategy. Static order submission strategies are presented in Section 7.2.1, while dynamic strategies are discussed in Section 7.2.2. The overview of all models discussed in this section is summarised in Table 7.1.

Table 7.1: Overview of previous work on order placement strategies. Each model is characterised by the trading problem it tries to solve, whether it is static or dynamic strategy and the market variables that it utilises.

Models	Problem Type			Strategy		Incorporated Variables
	I	II	III	Static	Dynamic	
Handa and Schwartz [41]	x			x		-
Parlour [76]		x		x		volume in the book
Foucault [32]		x		x		volatility
Hollifield et al. [49]		x		x		volatility, order quantity, volume at bid/ask, trading volume, time of day
Foucault et al. [33]		x		x		order arrival rate, spread
Nevmyvaka et al. [73]	x			x		order quantity, time of day, trading volume
Lillo [56]		x		x		volatility
Cohen et al. [19]		x			x	-
Harris [42]	x	x	x		x	volatility
Slive [87]		x			x	volatility, spread
Nevmyvaka et al. [72]	x				x	order quantity, spread, order imbalance, immediate cost, trading volume
Wang and Zhang [92]	x				x	order quantity, order imbalance

7.2.1 Static order submission strategies

As previously discussed, static order submission models consider the decision whether to submit a market order or to submit a limit order as a one-shot game where a trader can make their order decision only once. If the trader decides to submit a limit order, no additional change can be made to his order and it will stay in the order book until it is executed or the end of the trading period is reached. This formulation can be utilised to solve the problems of both liquidity traders [41, 73] and informed traders [19, 76, 32, 49, 33, 56]. The main difference between these two problems is that liquidity traders have to fill their order before the deadline; thus, if liquidity traders decide to submit a limit order and their orders are not executed, they have to submit a market order to execute the trade when their deadlines are approached. On the other hand, informed traders will submit market orders to fill the traders only when it is still profitable to do so.

Static strategies for liquidity traders

Consider the problem of liquidity traders who want to transact their orders before some specific deadlines. Normally, they can choose to submit their order using the following strategies: (i) submitting a market order at the beginning of the time period, (ii) submitting a market order at the end of the time period, and (iii) submitting a limit order at the beginning of the time period and a market order for unexecuted shares at the end of the time period.

Handa and Schwartz [41] analyse the profitability of the third strategy compared to the first one. The limit order strategy that they study is to submit a limit order placed l percent below the current price, where l is set to 0.5, 1, 2 and 3. The limit order is followed until it executes or until the last price in the trading window is reached. If the limit order does not execute during the trading window, the stock is purchased at the opening price on the day following the trading window. The experimentation results indicate that returns of limit order conditional on execution are positive, while returns of limit order conditional on nonexecution are negative. They also find that picking off risk is not a cost to limit

order traders, but that nonexecution is. Thus, it is more appropriate for liquidity trader to transact by the market order strategy, while traders who gain relatively little by trading at current prices (and who are willing to risk not executing) may prefer the limit order strategy.

Another study comparing the profitability of these three choices by Nevmyvaka, Kearns, Papan-dreou, and Sycara [73] suggests that the limit order strategy performs better than the market order strategy. Although this may seem to contradict [41], they study the limit strategy in finer detail. Specifically, they present a method to estimate return, risk, and risk-return profiles of each strategy from historical data, as well as a method to derive optimal pricing frontiers based on the trade-off between risk and return. Their quantitative method allows traders to optimally price their limit orders to minimise trading costs and control corresponding risks. The importance of a number of microstructure variables (e.g. order size, time window and liquidity) is also highlighted.

Static strategies for informed traders

Unlike liquidity traders, informed traders do not have a responsibility to fill the trade when the deadline is approached. Thus the decision these traders face is simply whether to trade aggressively by submitting a market order or to trade passively by placing a limit order. This static decision problem is usually formalised as an optimisation problem that considers the trade-off between the payoff associated with limit orders and the risk of nonexecution. On one hand, traders would prefer to place their orders very far from the best bid/ask price because this will increase their payoff. On the other hand, the larger the distance from the best price the larger the chance that the order will not be executed. Thus, in this setting, traders have to find the right trade-off between these two opposite choices in order to maximise the expected profit obtained from the trade. This section briefly reviews static order submission strategies for informed traders previously proposed in the literature.

Parlour [76] presents a model of the evolution of the limit order book. The optimal choice between submitting a limit order and a market order is characterised as a single-period optimisation model. The central intuition of her research is that each trader knows that his order will affect the order submission strategies of other traders who follow; thus, he take this effect into account, which in equilibrium, generate systematic patterns in prices and order placement strategies even without asymmetric information. Her study also suggests that both side of the order book are important in determining an agent's order choice.

Foucault [32] describes a game theoretical model of price formation and order submission decisions in a dynamic limit order market where traders arrive sequentially and choose to submit either a market order or a limit order with one-period life. His results indicate that (i) the proportion of limit orders in the order flow is positively related to asset volatility, (ii) the ratio of filled limit orders to total number of limit orders is negatively related to asset volatility, (iii) the proportion of limit orders is positively related to the average size of the spread, (iv) the increase in trading cost at the end of the trading day is negatively related to the level of competition between limit order traders and (v) the size of the sum of trading costs for buy and sell orders is maximised when the ratio of buy to sell orders, is equal to one.

Hollifield, Miller and Sandas [49] present empirical restrictions of a model of optimal order sub-

mission in limit order markets. A trader's optimal order submission depends on the trader's valuation of the asset and the trade-offs between order prices, execution probability, and picking off risks. The optimal order submission strategy is a monotonic function of a trader's valuation, characterised in terms of threshold valuations. The threshold valuations are functions of the order prices and the trader's subjective beliefs about the execution probabilities and picking off risks.

Foucault, Kadan and Kandel [33] propose a dynamic model of a limit order market populated by strategic liquidity traders of varying impatience who aim to optimise the trade-off between the cost of delayed execution and the cost of immediacy (the spread). The optimal order submission strategy of each trader is modelled as a single period optimisation problem. Under several simplifying assumptions, they derive the equilibrium order placement strategies. They find that the proportion of patient traders in the population and the order arrival rate are the key determinants of the limit order book dynamics. Traders submit aggressive limit orders, which improve current best quotes by large amounts, when the order arrival rate is low or when the proportion of patient traders is large. As a result, markets with a high proportion of patient traders or a small order arrival rate are more resilient. Also, a reduction in the tick size reduces market resiliency, and, in some case, increases the average spread. Their analysis also yields several testable predictions: (i) a positive relationship between inter-trader durations and market resiliency, (ii) a negative relationship between the order arrival rate and market resiliency, (iii) a joint decline of limit order aggressiveness and market resiliency at the end of the trading session and (iv) limit order traders submit more (less) aggressive orders when the spread is large if patient (impatient) traders dominate the trading population.

Lillo [56] considers the problem of the optimal limit order price for a financial asset in the framework of utility maximisation. The analytical solution of the problem gives insight into the origin of the recently empirically observed power law distribution of limit order prices. In the framework of the model, the most likely proximate cause of this power law is power law heterogeneity of traders' investment time horizons.

7.2.2 Dynamic order submission strategies

Unlike static order submission strategies, dynamic order submission strategies allow traders to monitor the changing market conditions and make changes to their order any time before the order expires or is executed. Empirically, traders change their order submission as market conditions change. They continuously monitor the market and make appropriate changes to their orders whenever necessary. For example, to reduce the execution risk, they may convert their limit orders to market orders when the demand for immediacy increases. They may also reprice or cancel their limit orders when the underlying value of the asset changes to manage the adverse selection cost. Hence, it is more appropriate to model this decision with dynamic strategies than with static strategies. This section presents the existing dynamic strategies previously proposed in the literature.

Cohen, Maier, Schwartz and Whitcomb [19] consider an order submission strategy as a dynamic optimisation problem. Traders in their model may seek to trade via limit order, trade with certainty via a market order, or not to trade at all. Their result demonstrates that transaction costs cause bid-ask

spreads to be an equilibrium property of financial markets since, with transaction costs, the execution probability of a limit order does go to unity as the order is placed infinitesimally close to the opposite market quote; thus, with certainty of execution at the opposite market quote, a "gravitational pull" that keeps the opposite quotes from being placed infinitesimally close to each other is generated. They also define an equilibrium market spread and illustrate that it is negatively related to the order arrival rate.

Harris [42] derives optimal dynamic order submission strategies for trading problems faced by three stylised traders: an uninformed liquidity trader, an informed trader and a value-motivated trader. Separate solutions are obtained for quote- and order-driven markets. These results suggest that traders are most aggressive when volatility is high and when their information advantages, if any, are large and decay quickly. Traders are patient when their deadlines are not pressing and when bid/ask spread are wide. The numerical results suggest that most traders should place limit orders close to the market when they trade. Although it may sometimes be optimal for risk neutral traders to place orders far from the market (when deadlines are distant or when private information will not be revealed soon), the expected additional benefits from this strategy are very small. If monitoring costs are high or if the trader is risk averse, distant order placement strategies will not be optimal. The only exception to this rule is for traders who believe that prices are mean-reverting. They may place limit orders far from the market to benefit if prices move far from fundamental values.

Nevmyvaka, Feng and Kearns [72] present the first large-scale empirical application of reinforcement learning to the problem of trade execution. In their problem, the goal is to sell (respectively, buy) V shares of a given stock within a fixed period of time in a manner that maximises the revenue received (respectively, minimises the capital spent). Their results indicate that introducing market variables into the model can greatly improve the execution result and reinforcement learning can indeed result in significant improvement over simpler forms of a single-period optimisation model.

Wang and Zhang [92] present dynamic focus strategies that incorporate a series of market orders of different volume into the limit order strategy and dynamically adjust their volume by monitoring state variables such as inventory and order book imbalance in real-time. The sigmoid function is suggested as the quantitative model to represent the relationship between the state variables and the volume to be adjusted. The empirical results indicate that the dynamic focus strategies can outperform the limit order strategy, which does not adopt dynamic volume adjustment.

Slive [87] derives the optimal dynamic order submission strategies of a trader in a limit order market who has the ability to actively monitor his order and use cancellations and order changes to mitigate the adverse selection and execution risks inherent in limit orders. His results suggest that the ability to implement a dynamic strategy has a large impact on the payoffs to submit limit orders and on limit order submission strategies. After calibrating the parameters to a stock on the Vancouver Stock Exchange, profits from limit order submission are 48% higher when implementing a dynamic strategy compared to a one-shot strategy. Cancellations and order changes are used to avoid adverse selection by moving orders when the underlying value changes. Order changes are used to mitigate execution risk by converting to a market order when the probability of execution declines.

7.3 Framework for an order placement strategy

This section presents a new framework for making order placement decisions which is general enough to solve all three trading problems mentioned above in a mean-variance optimisation framework by extending the model presented in [42]. This framework is based on the trade-off between the profit gained from better execution and the risk of non-execution. As a starting point, we will consider only a static strategy where traders can make decisions only once before the trade begins. After the order is submitted, no additional change can be made to the order until the trading period ends. In particular, we consider an order placement problem of a trader who wants to buy a particular instrument¹ within a time period T in order to maximise his utility function. In this situation, a trader has four possible choices:

1. Do nothing;
2. Execute the order at the beginning of the period at the current market price p_0^M ;
3. Execute the order at the end of the period at the closing price p_T^M ; or
4. Firstly, submit a limit order to execute a trade at a limit price p^L . If the order is not executed, the trader then either executes the order at the closing price or does nothing depending on his objective.

Although the trader has four choices, these choices can be represented by using only the last strategy. This is because the first and the third choices can be represented by a very low limit price so that the chance that the order will be executed is zero. The second choice can be represented by a limit price higher than the current market price so that the order will be executed immediately. To model these choices, the trader needs to specify his trading objective by defining two payoff functions: a function $U_E(p)$, that defines the payoff he will get when he executes the trade at a price p , and a function $U_{NE}(p)$, that defines the cost he needs to pay if his order is not executed when the asset price at the end of the period is p . Consequently, the payoff the trader will get from submitting a limit buy order at price level p^L is characterised by

$$U(p^L) = \begin{cases} U_E(p^L) & \text{,if the order is executed} \\ U_{NE}(p_T^M) & \text{,if the order is not executed} \end{cases} \quad (7.1)$$

This payoff is a random variable since its value depends on whether the submitted order will be executed or not as well as the price of the asset at the end of the period p_T^M , whose values are not known beforehand. Given the probability that the limit order at price p^L will be executed before the end of the period together with the distribution of the asset price at the end of the period, the expected payoff the trader might get can be computed from

$$E[U(p^L)] = P_E(p^L)U_E(p^L) + [1 - P_E(p^L)] \int_{-\infty}^{\infty} U_{NE}(p) f_{p_T^M|p^L}(p) dp, \quad (7.2)$$

¹ An order placement problem for a trader who want to sell a particular instrument can be formulated in a similar way

where $P_E(p^L)$ is the probability that the limit order at price p^L will be executed before the end of the period, and $f_{p_M|p^L}(\cdot)$ is the probability density function of the asset price at the end of the period given that the limit order at price level p^L is not executed, i.e., the asset price is never lower than or equal to p^L . Similarly, the variance of this utility function can be computed from

$$V[U(p^L)] = P_E(p^L)(U_E(p^L) - E[U(p^L)])^2 + [1 - P_E(p^L)] \int_{-\infty}^{\infty} (U_{NE}(p) - E[U(p^L)])^2 f_{p_M|p^L}(p) dp, \quad (7.3)$$

Inserting Equation (7.2) into (7.3), Equation (7.3) can be rewritten as

$$V[U(p^L)] = [1 - P_E(p^L)] \left[P_E(p^L) \left(U_E(p^L) - \int_{-\infty}^{\infty} U_{NE}(p) f_{p_M|p^L}(p) dp \right)^2 \right] + [1 - P_E(p^L)] \left[\int_{-\infty}^{\infty} [U_{NE}(p)]^2 f_{p_M|p^L}(p) dp - \left(\int_{-\infty}^{\infty} U_{NE}(p) f_{p_M|p^L}(p) dp \right)^2 \right] \quad (7.4)$$

To perform a mean-variance optimisation, a utility function of a trader who executes his trade by using a limit order at price level p^L could be defined as

$$U_O(p^L) = E[U(p^L)] - \lambda V[U(p^L)], \quad (7.5)$$

where λ is proportional to the trader's risk aversion, or inversely proportional to trader's aggressiveness. As an example, a value of $\lambda = 0$ indicates a trader who is concerned only about the profit gained from better execution while a value of $\lambda = 1$ indicates a trader who equally concerns about the profit gained from better execution and the risk of non-execution. Consequently, the optimal order placement strategy that balances the trade-off between the profit gained from limit orders and the risk of non-execution at the trader's specified level of risk aversion can be determined by maximising the above utility function and can be defined as

$$\begin{aligned} \hat{p} &= \operatorname{argmax}_{p^L} U_O(p^L), \\ &= \operatorname{argmax}_{p^L} E[U(p^L)] - \lambda V[U(p^L)]. \end{aligned} \quad (7.6)$$

By specifying the form of these two utility functions, this framework can be utilised to solve all three trading problems mentioned in the beginning of this chapter. In particular, the problem of liquidity traders who need to transact shares before a deadline can be modelled by setting $U_{NE}(p) = U_E(p)$, so that the cost that a trader needs to pay if his limit order is not executed is equal to the cost of buying at the price at the end of the period. The utility function of informed traders and value motivated traders can be modelled by setting $U_{NE}(p) = \max\{U_E(p), U_{NT}\}$, where U_{NT} is a utility gained from not trading, so that they will trade at the end of the period only when it is profitable to do so. To completely specify this framework, one also needs to specify a model of the probability that the limit order at price level p^L

will be executed before the end of the period, as well as a probability density function of the price at the end of the period given that the limit order submitted at price level p^L is not executed. In the rest of this chapter, we will utilise the price fluctuation model developed in Chapter 5 to model the probability of execution, while utilise the ARMA-GARCH model to model the probability density function of the price at the end of the period. More detail about this will be discussed in the next section where we derive the optimal order placement strategy for liquidity traders.

7.4 Order placement strategy for liquidity traders

This section derives an order placement strategy for liquidity traders who need to transact their orders before some specified deadline from the framework discussed in the previous section. Particularly, we consider a problem of how a trader who want to buy shares should submit his order to execute the trade before the deadline T . In this setting, a trader has three possible choices:

1. Execute the order at the beginning of the trading period, at the current market price p_0^M ;
2. Execute the order at the end of the trading period, at the future closing price p_T^M ; or
3. Submit a limit order at price level p^L , at the beginning of the trading period. If the order is not executed, the trader then executes the trade at the future closing price p_T^M .

In all these cases, the order is guaranteed to have been executed by the end of the period but the money spent to open this position will be different. The objective of this section is to determine the best way to execute this order to get the most favourable price at a specified risk aversion parameter. To achieve this, Section 7.4.1 firstly derives the model for optimising this decision based on the framework discussed in the previous section. We then describe three different approaches for modelling the execution probability and the probability density of the asset price at the end of the trading period. Particularly, Section 7.4.2 discusses the unconditional model implied from an arithmetic Brownian model. The unconditional empirical model obtained from density estimation is described in Section 7.4.3, while the conditional empirical model using ACD and ARMA-GARCH model is presented in Section 7.4.4. Finally, the results obtained from applying these model to make trading decision are analysed in Section 7.4.5.

7.4.1 Order placement model for liquidity traders

To utilise the proposed framework to make order placement decisions for liquidity traders, one needs to specify a utility function $U_E(p)$ that defines the payoff the trader will get from executing a trade at price p , a utility function $U_{NE}(p)$ that defines the cost the trader will pay if his order is not executed and the price at the end of the period is p , a probability model $P_E(p)$ that describes the probability that the order submitted at price p will be executed before the end of the period, and a probability density function $f_{p_T^M | p^L, p_0^M}(p_T^M)$ that describes the probability that the closing price will be p_T^M , if the limit order submitted at price p^L is not executed. In this setting, we specify the payoff that the trader will get from executing his order at price p^L as the difference between the current best ask price and the execution

price, or equivalently, we set

$$U_E(p^L) = p_0^M - p^L, \quad (7.7)$$

where p_0^M is the market price at the beginning of the trading period. Similarly, the payoff that the trader will get if his order is not executed, when the price at the end of the period is p_T^M , is defined as

$$U_{NE}(p_T^M) = p_0^M - p_T^M, \quad (7.8)$$

which is basically the profit gained from executing the order at the end of the trading period. Although the payoff the trader gets from executing his order at limit price p^L is always greater than zero, as the limit price p^L must be lower than p_0^M , the payoff that the trader will get if his order is not executed can be both positive and negative. In fact, this payoff will be negative if the asset price increases at the end of the trading period, while it will be positive if the asset price decreases at the end of the period. Consequently, when the limit order submitted by the trader is not executed, the trader will still gain price improvement if the asset price decreases and he will suffer a loss only when the asset price increases.

The probability that the limit order at each price level will be executed and the distribution of the asset price at the end of the trading period, given that the limit order place at price level p^L is not executed, can be modelled by several approaches. In this section, we will discuss three different approaches for modelling these two distributions, which are the distribution implied by an arithmetic Brownian motion, the unconditional model that utilises the empirical distribution, and the conditional model that uses the ACD model and ARMA-GARCH model.

7.4.2 Unconditional model implied by the arithmetic Brownian motion

The first and only theoretical model considered here is the model implied by an arithmetic Brownian motion. In this situation, the asset price is assumed to follow the arithmetic Brownian motion and a limit order is assumed to be executed when the asset price hits or crosses the limit price. Particularly, the asset price p_t is assumed to follow

$$dp_t = \sigma dW_t + \mu dt, \quad (7.9)$$

where μ is the instantaneous drift, σ is the standard deviation, and W_t is a Wiener process. In the context of a buy problem, p_t is best thought of as the best ask price and the limit order submitted at price level p^L will be executed before the end of the trading period T only when $p_t \leq p^L$ for some time t , $0 \leq t \leq T$. As discussed in Section 5.4.1, the probability that the limit buy order submitted at price level $p^L < p_0$ will be executed before the end of the period T can be obtained from

$$P_E(p^L|p_0) = \Phi\left(\frac{p^L - p_0 - \mu T}{\sigma\sqrt{T}}\right) + \exp\left(\frac{2\mu(p^L - p_0)}{\sigma^2}\right) \Phi\left(\frac{p^L - p_0 + \mu T}{\sigma\sqrt{T}}\right), \quad (7.10)$$

where $\Phi(\cdot)$ is the cumulative distribution function of a standard Normal distribution. When there are no constraints, the distribution of the asset price at the end of the trading period is simply a Normal distribution with mean μT and variance $\sigma^2 T$. However, the condition that the limit order at price level p^L is not executed constrains p_t to be higher than p^L for all $0 < t < T$. This means that the distribution

of p_T is not simply a Normal distribution truncated on the left at p^L but is a more complicated expression given by (see [44] page 165)

$$f_{p_T|p^L,p_0}(p_T) = \frac{\phi\left(\frac{p_0-p_T+\mu T}{\sigma\sqrt{T}}\right) - \exp\left(\frac{2\mu(p^L-p_0)}{\sigma^2}\right)\phi\left(\frac{2p^L-p_0-p_T+\mu T}{\sigma\sqrt{T}}\right)}{\sigma\sqrt{T}(1-P_E(p^L|p_0))}, \quad (7.11)$$

where $\phi(\cdot)$ is the probability density function of a standard Normal distribution. To gain more insight into the property of this distribution, Figure 7.1 plots this distribution in several parameter settings. The results displayed in the first column indicates that the distribution of asset prices at the end of the trading period given no execution of limit order at price level p^L is generally right-skewed and tends to shift upward to the right when p^L increases. This suggests that this distribution typically has higher mean and lower variance as p^L increases. The results displayed in the second and third columns convey similar outcomes as this distribution tends to shift downward to the right when the drift parameter and the volatility parameter increases. This suggests that this distribution generally has higher mean and higher variance when the drift and volatility parameter of the arithmetic Brownian motion increase.

By inserting Equation (7.10) and (7.11) into Equation (7.2) and (7.4), one can compute the expectation and the variance of the payoff the trader will get from executing his trade using a limit order at each price level under the arithmetic Brownian motion assumption. Consequently, the optimal order placement strategy for a specified level of risk aversion parameter can be derived accordingly by finding the limit order price that maximises the utility function described in Equation (7.5). To gain more insight into the optimal order placement strategy generated from this model, Figure 7.2 displays the expectation and the variance of this payoff function at several parameter settings. The result displayed in the first row of this figure indicates that this expectation is a monotonic decreasing function when the drift parameter is negative, a straight horizontal line when the drift parameter is zero, and a monotonic increasing function when the drift parameter is positive. This indicates that the optimal order placement strategy of a trader whose only concern is the expected profit gained from limit order trading depends heavily on the drift parameter of the arithmetic Brownian motion in the sense that it is always optimal to execute the order immediately at the beginning of the trading period when the drift parameter is positive, while it is

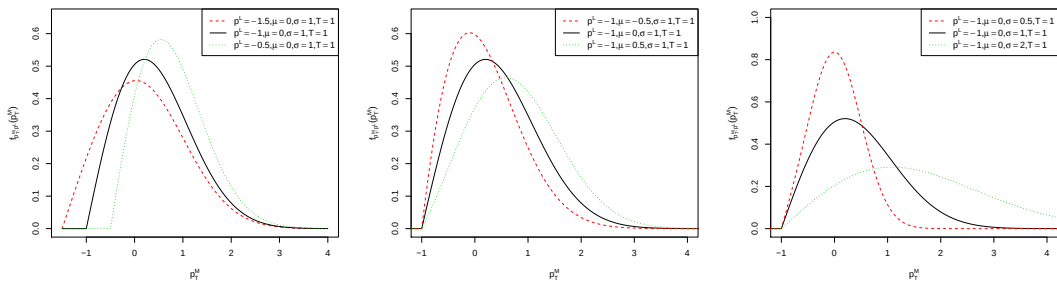


Figure 7.1: The probability density function of the asset price at the end of the trading period T given no execution of the limit order at price level p^L when the asset price is assumed to follow the arithmetic Brownian motion with the drift parameter equal to μ and the volatility parameter equal to σ at several parameter settings.

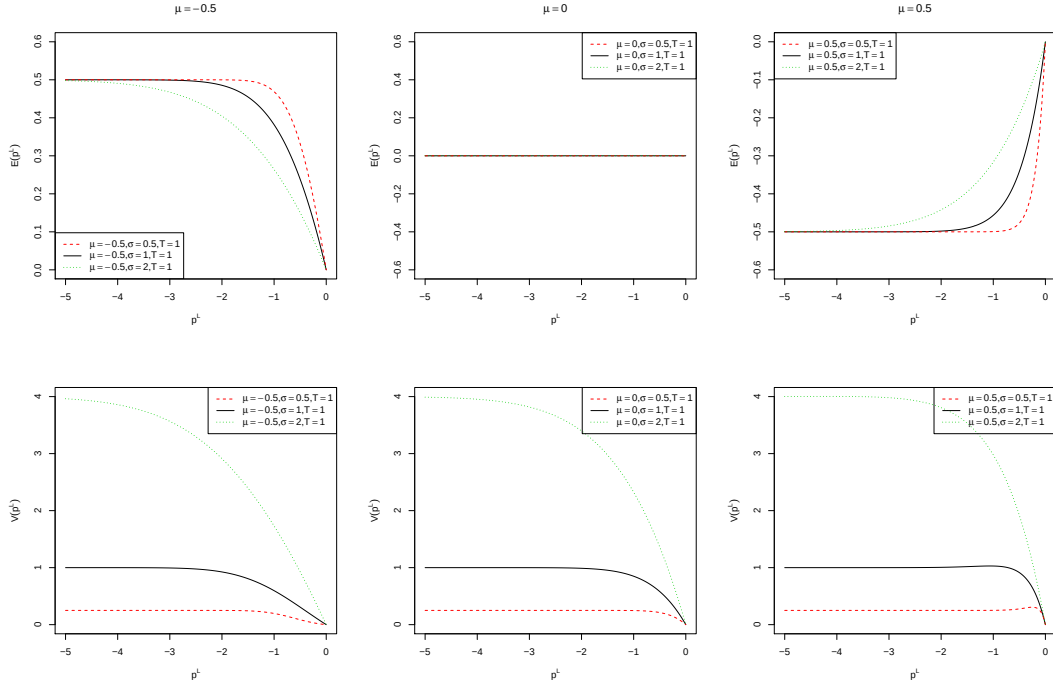


Figure 7.2: The expectation (top row) and the variance (bottom row) of the profit that liquidity traders will get from executing their trade using limit order at price level p^L when the asset price is assumed to follow the arithmetic Brownian motion with the drift parameter equal to μ and the volatility parameter equal to σ at several parameter settings.

always optimal to execute the order at the end of the period when the drift is negative. When the drift parameter is zero, all trading strategies are optimal since the expectation of this payoff is the same for all price levels. Additionally, the variance of this payoff function, displayed in the second row of Figure 7.2 indicates that the variance of this payoff is always a monotonic decreasing function with a minimum at zero. This suggests that it is always optimal to execute the trade immediately at the beginning of the trading period if the trader's objective is to minimise the risk of non-execution.

To analyse the optimal strategy of a trader who is concerned about the trade-off between the profit gained from limit order trading and the risk of non-execution, we plot the combined utility function at three different risk aversion levels when the drift parameter is negative in Figure 7.3². The result indicates that the combined utility function can be monotonic increasing, monotonic decreasing as well as unimodal depending on the parameters of the model. Unlike the optimal strategy of the trader whose only concern is the expected payoff or the risk of non-execution, the optimal strategy in this case also involves the use of limit orders especially when the combined utility function is unimodal. This also illustrates that the optimal strategy depends heavily on the parameters of the model, as the optimal limit price tends to move towards zero, which results in immediate execution, when trader's risk aversion level or the volatility parameter of the arithmetic Brownian motion increases.

²We only analyse the case when the drift parameter is negative since it is always optimal to execute the trade immediately at the beginning of the trading period when the drift parameter is positive.

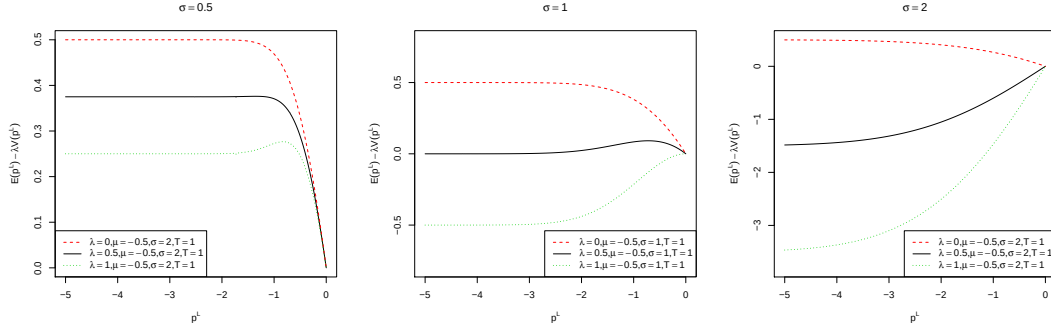


Figure 7.3: The utility function that liquidity traders will get from executing their trade using limit order at price level p^L at three different risk aversion levels, $\lambda = 0.1, 0.5, 1.0$, when the asset price is assumed to follow the arithmetic Brownian motion with drift parameter equal to μ and the volatility parameter equal to σ at several parameter settings.

7.4.3 Empirical unconditional model using density estimation

Since the results analysed in Section 5.4.1 indicates that the probability execution model implied by the arithmetic Brownian motion does not fit very well with the empirical distribution, it might be more appropriate to make order placement decision from the empirical distributions. To achieve this, this section presents a method for estimating the unconditional distribution of the execution probability and the asset price at the end of the trading period given no execution from the history of asset price. In particular, we estimated the probability that the limit order submitted at each price level will be executed before the end of the trading period is estimated from the empirical distribution of price fluctuations as discussed in Section 5.2, while the probability density function of future closing price given no execution will be estimated from the empirical distribution of asset returns at the end of the trading period.

Before discussing a method for modelling these two distributions from asset price history, we will firstly review the relationship between price fluctuations and the probability of execution as discussed in Section 5.2. Define a buy price fluctuation during time T , denoted by M_T^B , as the difference between the initial price level and the lowest price level reached during the trading period T , or equivalently

$$M_T^B = \sup\{p_0 - p_t; 0 \leq t \leq T\}. \quad (7.12)$$

Accordingly, the probability that a limit buy order submitted at price level $p^L < p_0$ will be executed before the end of the period T can be estimated from

$$\begin{aligned} P_E(p^L|p_0) &= \Pr\{\sup\{p_0 - p_t; 0 \leq t \leq T\} \geq p_0 - p^L\}, \\ &= \Pr\{M_T^B \geq p_0 - p^L\}, \\ &= 1 - F_{M_T^B}(p_0 - p^L), \end{aligned} \quad (7.13)$$

where $F_{M_T^B}(\cdot)$ is a cumulative distribution function of the buy price fluctuations during time period T . Given a history of buy price fluctuations $(\Delta_1, \dots, \Delta_2)$, this unconditional cumulative distribution

function can be estimated from

$$F_{M_T^E}(\Delta) = \frac{1}{N} \sum_{i=1}^N I\{\Delta_i \leq \Delta\}, \quad (7.14)$$

where $I\{\cdot\}$ is an indicator function which is equal to one when the enclosed expression is true, while it will be zero when the enclosed expression is false. Inserting Equation (7.14) into (7.13), the probability that the limit order at price level p^L will be executed before the end of the period can be estimated from

$$P_E(p^L|p_0) = 1 - \frac{1}{N} \sum_{i=1}^N I\{\Delta_i < p_0 - p^L\}. \quad (7.15)$$

Let us now consider a method to estimate the probability density function of the asset price at the end of the trading period given no execution of the limit order at price level p^L from the empirical distribution of asset returns at the end of the period. Given a history of asset price dynamics sampling every T unit time $(p_0, \dots, p_N)$, the unconditional empirical distribution of asset returns at the end of the trading period T can be estimated from

$$f_{r_T}(r) = \frac{1}{N} \sum_{i=1}^N I\{p_i - p_{i-1} = r\}. \quad (7.16)$$

Consequently, under the assumption that the asset price at the end of the period is independent of the execution of limit order at price level p^L , the distribution of asset prices at the end of the period can be estimated from

$$f_{p_T^M|p_0^M}(p_T^M) = f_{r_T}(p_T^M - p_0^M) = \frac{1}{N} \sum_{i=1}^N I\{p_i - p_{i-1} = p_T^M - p_0^M\}. \quad (7.17)$$

However, this assumption is generally violated in most situations since the condition that a limit order at price level p^L is not executed constrains the asset price to be higher than p^L for all time. This suggests that it might not be appropriate to estimate the probability density function of the asset price at the end of the trading period given no execution of limit order at price level p^L using the above equation. To improve this estimation, we choose to model this distribution using a left-truncated version of the above distribution which can be computed from

$$\begin{aligned} f_{p_T^M|p^L, p_0^M}(p_T^M) &= f_{r_T}(p_T^M - p_0^M | p_T^M \geq p^L) \\ &= f_{r_T}(p_T^M - p_0^M | p_T^M - p_0^M \geq p^L - p_0^M) \\ &= \frac{\sum_{i=1}^N I\{p_i - p_{i-1} = p_T^M - p_0^M\}}{\sum_{i=1}^N I\{p_i - p_{i-1} \geq p^L - p_0^M\}}. \end{aligned} \quad (7.18)$$

Since the distribution considered in this section is discrete in nature, the expectation of the payoff function in Equation (7.2) is changed to

$$E[U(p^L)] = P_E(p^L)U_E(p^L) + [1 - P_E(p^L)] \sum_{i=-\infty}^{\infty} U_{NE}(p_i^L + i\delta) f_{p_T^M|p^L, p_0^M}(p_i^L + i\delta), \quad (7.19)$$

where δ is the tick size of the instrument considered. Similarly, the variance of this payoff in Equation (7.4) is changed to

$$\begin{aligned} V[U(p^L)] = & [1 - P_E(p^L)] \left[P_E(p^L) \left(U_E(p^L) - \sum_{i=-\infty}^{\infty} U_{NE}(p^L + i\delta) f_{p_T^M|p^L, p_0^M}(p^L + i\delta) \right)^2 \right] \\ & + [1 - P_E(p^L)] \left[\sum_{i=-\infty}^{\infty} [U_{NE}(p^L + i\delta)]^2 f_{p_T^M|p^L, p_0^M}(p^L + i\delta) \right] \\ & - [1 - P_E(p^L)] \left[\sum_{i=-\infty}^{\infty} U_{NE}(p^L + i\delta) f_{p_T^M|p^L, p_0^M}(p^L + i\delta) \right]^2. \end{aligned} \quad (7.20)$$

Consequently, one can determine the optimal order placement strategy for a liquidity trader with a specified risk aversion level λ by finding a limit order p^L that maximises the combined utility function

$$\hat{p}^L = \underset{p^L}{\operatorname{argmax}} (E[U(p^L)] - \lambda V[U(p^L)]). \quad (7.21)$$

7.4.4 Empirical conditional model using ACD and ARMA-GARCH models

Whilst the models considered in the previous two sections are mainly focused on the unconditional distribution of the probability that the limit order will be executed before the deadline and the unconditional distribution of the asset price at the end of the trading period, it might be more useful to consider conditional models of these two distributions so that the optimal order placement strategy can adapt according to current market situations rather than fixing at a specified price level. To achieve this, this section presents a method for estimating the conditional distribution of the execution probability using the Autoregressive Condition Duration (ACD) model as described in Section 5.5.3, while modelling the distribution of asset price at the end of the trading period using the Autoregressive Moving Average - Generalised Autoregressive Conditional Heteroskedasticity (ARMA-GARCH) model.

Let $(\Delta_1, \dots, \Delta_N)$ be a sequence of price fluctuations during the considered time period. The ACD(p, q) model for price fluctuation, as discussed in Section 5.5.3 and 5.6, can be specified by

$$\begin{aligned} \Delta_i &= \hat{\Delta}_i s(t_i), \\ \hat{\Delta}_i &= \psi_i \epsilon_i, \\ \psi_i &= \Psi(\hat{\Delta}_{i-1}, \dots, \hat{\Delta}_{i-p}, \psi_{i-1}, \dots, \psi_{i-q}), \end{aligned}$$

where $\hat{\Delta}_i$ is the seasonally adjusted price fluctuations at the i -th time step, $s(t_i)$ is the seasonality component at the i -th time step, ψ_i is the conditional expectation of price fluctuations at the i -th time step, $\{\epsilon_i\}$ is an independent and identically distributed white noise with unit mean and finite variance, and $\Psi(\cdot)$ is the conditional mean function. If $\{\epsilon_i\}$ has a cumulative distribution function $F_\epsilon(\cdot)$, the conditional distribution of price fluctuations at the i -th time step, when the tick size of the considered instrument is δ , will be given by

$$f_{ACD}(\Delta_i | \mathcal{F}_{i-1}) = F_\epsilon\left(\frac{\Delta_i + \delta}{\psi_i s(t_i)}\right) - F_\epsilon\left(\frac{\Delta_i}{\psi_i s(t_i)}\right), \quad (7.22)$$

and the probability that the limit order at price level p_i^L , submitted at the i -th time step will be executed before the end of the trading period can be computed from

$$\begin{aligned} P_E(p_i^L | p_{i-1}) &= 1 - \Pr \{ \Delta_i < p_{i-1} - p_i^L \}, \\ &= 1 - \sum_{i=-\infty}^{(p_{i-1} - p_i^L)/\delta - 1} f_{ACD}(i\delta), \\ &= 1 - F_\epsilon \left(\frac{p_{i-1} - p_i^L}{\psi_i s(t_i)} \right). \end{aligned} \quad (7.23)$$

In the rest of this section, we will assume that the conditional mean function has the form of the ABAMACD model which can be specified by

$$\begin{aligned} \psi_i^{\delta_1} &= \omega + \sum_{j=1}^p \alpha_j \psi_{i-j}^{\delta_1} (|\epsilon_{i-j} - b| + c_j (\epsilon_{i-j} - b))^{\delta_2} \\ &\quad + \sum_{j=1}^p \lambda_j (|\epsilon_{i-j} - b| + c_j (\epsilon_{i-j} - b))^{\delta_2} + \sum_{j=1}^q \beta_j \psi_{i-j}^{\delta_1}, \end{aligned} \quad (7.24)$$

where $\theta_\Psi = (\delta_1, \delta_2, b, c_1, \dots, c_p, \alpha_1, \dots, \alpha_p, \lambda_1, \dots, \lambda_p, \beta_1, \dots, \beta_q)$ are the parameters of the model. The seasonality component at the i -th time step will be specified by a time-of-day equation

$$s(t_i) = \nu_1 I_1(t_i) + \nu_2 I_2(t_i) + \dots + \nu_s I_s(t_i), \quad (7.25)$$

where $\theta_s = (\nu_1, \dots, \nu_s)$ are the parameters to be estimated and $I_k(t_i)$ is an indicator function whose value can be either one or zero indicating whether the time t_i is in a particular time interval of the day or not. Additionally, the independent and identically distributed white noise $\{\epsilon_i\}$ will be assumed to follow the unit generalised gamma distribution whose cumulative density function has the form

$$F_{GG}(\epsilon) = \frac{\gamma(\lambda, [\epsilon \Gamma(\lambda + 1/\kappa) / \Gamma(\lambda)]^\kappa)}{\Gamma(\lambda)}, \quad (7.26)$$

where $\theta_\epsilon = (\kappa, \lambda)$ are parameters of the generalised gamma distribution, $\gamma(\lambda, x) = \int_0^x t^{\lambda-1} e^{-t} dt$ is a lower incomplete gamma function, and $\Gamma(\lambda) = \int_0^\infty t^{\lambda-1} e^{-t} dt$ is a gamma function. Consequently, given a time-series of price fluctuations $(\Delta_1, \dots, \Delta_N)$ the parameters of this model can be estimated by maximising the log-likelihood of the probability density function described in Equation (7.22), or, equivalently, the maximum likelihood estimator $\hat{\theta} = (\hat{\theta}_\Psi, \hat{\theta}_s, \hat{\theta}_\epsilon)$ of this model can be obtained from

$$\begin{aligned} \hat{\theta} &= \underset{\theta \in (\theta_\Psi, \theta_s, \theta_\epsilon)}{\operatorname{argmax}} \sum_{i=p+1}^N \ln f_{ACD}(\Delta_i | \mathcal{F}_{i-1}), \\ &= \underset{\theta \in (\theta_\Psi, \theta_s, \theta_\epsilon)}{\operatorname{argmax}} \sum_{i=p+1}^N \ln \left(F_{GG} \left(\frac{\Delta_i + \delta}{\psi_i s(t_i)} \right) - F_{GG} \left(\frac{\Delta_i}{\psi_i s(t_i)} \right) \right), \end{aligned} \quad (7.27)$$

where ψ_i are computed from Equation (7.24) using the observations $\Delta_1, \dots, \Delta_N$ with $\psi_i = \hat{\Delta}$ for all $i \leq 0$ when $\hat{\Delta}$ is the sample mean of $(\Delta_1, \dots, \Delta_N)$.

To model the probability density function of the asset price at the end of the i -th time step given no execution of limit order at price level p_i^L , we will utilise the ARMA-GARCH model to model the return at the i -th time step, and, then, estimate the density of the asset price at the end of the i -th time step given no execution of limit order at price level p^L from the density of the return at the i -th time step, as discussed in the previous section. Particularly, let $(p_0, \dots, p_N)$ be a series of asset price sampling every T unit time. We model the return of the i -th time step, $r_i = p_i - p_{i-1}$, using the ARMA(p,q)-GARCH(r,s) model of the form

$$\begin{aligned} r_i &= \hat{r}_i + s_r(t_i), \\ \hat{r}_i &= \omega + \sum_{j=1}^p \alpha_j r_{i-j} + \sum_{j=1}^q \beta_j z_{i-j} + z_i, \\ \sigma_i^2 &= \gamma + \sum_{j=1}^r \theta_j z_{i-j}^2 + \sum_{j=1}^s \lambda_j \sigma_{i-j}^2, \\ z_i &= \sigma_i \epsilon_i, \end{aligned} \quad (7.28)$$

where $\theta_a = (\omega, \gamma, \alpha_1, \dots, \alpha_p, \beta_1, \dots, \beta_q, \theta_1, \dots, \theta_r, \lambda_1, \dots, \lambda_s)$ are the parameters of the model, $\{\epsilon_i\}$ is an independent and identically distributed white noise with zero mean and unit variance, $\hat{r}_i$ is the seasonality adjusted return at the i -th time step, and $s_r(t_i)$ is the seasonality component at the i -th time step which has the form

$$s_r(t_i) = \nu_1 I_1(t_i) + \nu_2 I_2(t_i) + \dots + \nu_s I_s(t_i), \quad (7.29)$$

where $\theta_s = (\nu_1, \dots, \nu_s)$ are the parameters to be estimated and $I_k(t_i)$ is an indicator function whose value can be either one or zero indicating whether the time t_i is in a particular time interval of the day or not, similar to the seasonality component of the ACD model. If the white noise $\{\epsilon_i\}$ has a probability density function $f_\epsilon(\cdot)$, the conditional distribution of the asset return at the i -th time step can be computed from

$$\begin{aligned} f_{r_i}(r_i | \mathcal{F}_{i-1}) &= \frac{1}{\sigma_i} f_\epsilon \left(\frac{z_i}{\sigma_i} \right), \\ &= \frac{1}{\sigma_i} f_\epsilon \left(\frac{r_i - (\omega + \sum_{j=1}^p \alpha_j r_{i-j} + \sum_{j=1}^q \beta_j z_{i-j} + s(t_i))}{\sigma_i} \right). \end{aligned} \quad (7.30)$$

Consequently, under the assumption that the asset price at the end of the period is independent of the execution of limit orders at price level p^L , the distribution of the asset price at the end of the i -th time step given no execution of limit order at price level p_i^L can be estimated from

$$\begin{aligned} f_{p_i}(p_i | \mathcal{F}_{i-1}) &= f_{r_i}(p_i - p_{i-1} | \mathcal{F}_{i-1}) \\ &= \frac{1}{\sigma_i} f_\epsilon \left(\frac{p_i - p_{i-1} - (\omega + \sum_{j=1}^p \alpha_j r_{i-j} + \sum_{j=1}^q \beta_j z_{i-j} + s(t_i))}{\sigma_i} \right), \end{aligned} \quad (7.31)$$

while the probability density function of the asset price at the end of the i -th time step given no execution

of limit order at price level p_i^L , using the left-truncated version of the distribution above can be computed from

$$\begin{aligned}
 f_{p_i|p^L}(p_i|\mathcal{F}_{i-1}) &= f_{r_i}(p_i - p_{i-1}|p_i \geq p^L, \mathcal{F}_{i-1}), \\
 &= \frac{f_{r_i}(p_i - p_{i-1}|\mathcal{F}_{i-1})}{1 - \int_{-\infty}^{p^L} f_{r_i}(p - p_{i-1}|\mathcal{F}_{i-1}) dp}, \\
 &= \frac{f_{\epsilon}\left(\left(p_i - p_{i-1} - \left[\omega + \sum_{j=1}^p \alpha_j r_{i-j} + \sum_{j=1}^q \beta_j z_{i-j} + s(t_i)\right]\right) / \sigma_i\right)}{\sigma_i \left[1 - F_{\epsilon}\left(\left(p_i - p_{i-1} - \left[\omega + \sum_{j=1}^p \alpha_j r_{i-j} + \sum_{j=1}^q \beta_j z_{i-j} + s(t_i)\right]\right) / \sigma_i\right)\right]}, \tag{7.32}
 \end{aligned}$$

where $F_{\epsilon}(\cdot)$ is the cumulative distribution function of the independent and identically distributed white noise $\{\epsilon_i\}$. In the rest of this section, we will assume that distribution of the white noise $\{\epsilon_i\}$ follows a standard Normal distribution whose probability density function and cumulative distribution function are

$$f_{NN}(\epsilon) = \frac{1}{\sqrt{2\pi}} \exp\left(-\frac{\epsilon^2}{2}\right) \text{ and } F_{NN}(\epsilon) = \frac{1}{2} \left[1 + \operatorname{erf}\left(\frac{\epsilon}{\sqrt{2}}\right)\right], \tag{7.33}$$

where $\operatorname{erf}(x) = \frac{2}{\sqrt{\pi}} \int_0^x e^{-t^2} dt$ is the Gauss error function. Consequently, given a sequence of asset prices $p_0, \dots, p_N$, the parameter of this ARMA-GARCH model can be estimated by maximising the log-likelihood implied by the probability density function described in Equation (7.31), or equivalently, the maximum likelihood estimator $\hat{\theta} = (\theta_a, \theta_s)$ of this ARMA(p, q)-GARCH(r, s) model can be computed from

$$\hat{\theta} = \operatorname{argmax}_{\theta \in (\theta_a, \theta_s, \theta_{\epsilon})} \sum_{i=p+1}^N \left[\ln f_{NN}\left(\frac{z_i}{\sigma_i}\right) - \ln \sigma_i \right], \tag{7.34}$$

where z_i are computed from Equation (7.28) using the observation $p_0, \dots, p_N$ with $z_i = \mathbb{E}z_i = 0$ for $i = \min(p - q + 1, p), \dots, p$. Since the distribution considered in this section is discrete, the expectation of the payoff function for executing the trade using limit orders at price p_i^L at the i -th time step is

$$\begin{aligned}
 E[U(p_i^L)] &= P_E(p_i^L|p_{i-1})U_E(p_i^L) \\
 &\quad + [1 - P_E(p_i^L|p_{i-1})] \sum_{j=-\infty}^{\infty} U_{NE}(p_i^L + j\delta) f_{p_i|p_i^L}(p_i^L + j\delta), \tag{7.35}
 \end{aligned}$$

where δ is the tick size of the instrument considered. Similarly, the variance of this payoff at the i -th

time step is characterised by

$$\begin{aligned}
 V[U(p_i^L)] = & [1 - P_E(p_i^L | p_{i-1})] \left[P_E(p_i^L | p_{i-1}) \left(U_E(p_i^L) - \sum_{j=-\infty}^{\infty} U_{NE}(p_i^L + j\delta) f_{p_i | p_i^L}(p_i^L + j\delta) \right)^2 \right] \\
 & + [1 - P_E(p_i^L | p_{i-1})] \left[\sum_{j=-\infty}^{\infty} [U_{NE}(p_i^L + j\delta)]^2 f_{p_i | p_i^L}(p_i^L + j\delta) \right] \\
 & - [1 - P_E(p_i^L | p_{i-1})] \left[\sum_{j=-\infty}^{\infty} U_{NE}(p_i^L + j\delta) f_{p_i | p_i^L}(p_i^L + j\delta) \right]^2.
 \end{aligned} \tag{7.36}$$

Consequently, one can determine the optimal order placement strategy for a liquidity trader with a specified risk aversion level λ at the i -th time step by finding a limit order p^L that maximises the combined utility function

$$\hat{p}_i^L = \underset{p_i^L}{\operatorname{argmax}} [E[U(p_i^L)] - \lambda V[U(p_i^L)]]. \tag{7.37}$$

7.4.5 Experimentation results

This section investigates the performance of applying the proposed model to make order placement decisions for liquidity traders by utilising the historical data from the Multi Commodity Exchange of India (MCX) and the New York Stock Exchange (NYSE) as described in Section 5.3.1. In particular, we focus here on six instruments, which are the gold, silver, and natural gas futures contracts traded at the MCX and the GE, IBM and Microsoft stock traded at the NYSE. To measure the performance of the proposed order placement model, the trading period, T , for the instruments in the MCX is set to five minutes, while the trading period for the instruments in the NYSE is set to ten minutes. To reveal any bias in our results, we test our model on both a buy and a sell problem. We also vary the risk aversion parameters at several levels to analyse the effect of this parameter on the trading performance.

Since the order placement strategy obtained from unconditional models is static in the sense that they will execute the trade at the same limit order price in all situations, we will not analyse their results directly but will represent them using the static strategy that has the best performance in each dataset. However, we will replace some components of the conditional model discussed in Section 7.4.4 with the corresponding component of the unconditional model discussed in Section 7.4.3 with the aim of identifying the improvement gained from the conditional model. In particular, we will perform the experiment with three order placement models which are: i) the ACD-ARMA model that models the probability of execution and the density of the asset price at the end of the trading period using the ACD model and ARMA-GARCH model, respectively, ii) the ACD-DENSITY model that models the probability of execution using the ACD model, while modeling the density of future closing prices using density estimation, and iii) DENSITY-ARMA that utilises the density estimation to model the probability of execution but utilises the ARMA-GARCH model to model the density of the future closing prices. Additionally, since the probability density of the asset price at the end of the trading period given no execution of limit order can be estimated under both independent and truncated assumptions, for each of the three models,

we also estimate this distribution under both assumptions. As a result, this section will investigate six order placement models as summarised in Table 7.2. For each of these models, we firstly estimate its parameters from the first 75% of the dataset using the method described in Section 7.4.3 and 7.4.4. We then perform an experiment to find the risk aversion level that produces the best result in the first 75% of the dataset, and utilise the best risk aversion level to perform trading in the last 25% of the dataset. Finally, we compare the performance of the best trading strategy obtained from each model in order to determine the best model for making order placement decisions.

Trading with no market variables

Let us firstly analyse the results obtained from applying the ACD and ARMA-GARCH model without any market variables. Table 7.3 displays the trading results for the IBM stock in NYSE market using the above models at several risk aversion levels. The results indicate that the risk aversion level that provide the best profit is not zero, but generally higher than that. However, the result clearly confirms the important of the risk aversion parameters in controlling the risk of non-execution as the variance of the profit gained from using the strategy is generally lower when the risk aversion parameter is higher.

Table 7.4 and Table 7.5 report the performance of the best risk aversion level for each of the six models together with the performance of the static strategy that always executes the trade immediately at the beginning of the period, the static strategy that always executes the trade at the end of the trading period and the static strategy that always executes the trade at the best level. The result indicates that all of the six models beat the strategy that executes the trade at the beginning and the end of the trading period in all situations both in the training and testing dataset. Additionally, the best of these six models can also beat the best static strategy in all instruments in the MCX market, while, in the NYSE market, it can beat the static strategy only in the IBM stock. Although our proposed model cannot beat the best static strategy in all cases, the improvement gained from our model when it does beat the best static strategy is significant and ranges from 0.1% to 6.4% in the training dataset and 0.1% to 14.2% in the testing dataset. However, no clear winning models can be identified since the best models vary considerably from case to case. This suggests that it is more appropriate to try all possible models and select the best performing one to make trading decisions.

To compare the performance of these models, we calculate the profit gained (in number of ticks) from these strategies over immediately executing the trade at the beginning of the period averaged over all instruments in the same markets. The result reported in Table 7.6 indicates that the ACD-ARMA model with independent assumption seems to be the best performing model with 26.25% and 15.72%

Table 7.2: List of models for making order placement decisions.

Short name	Model
T-ACD-ARMA	The ACD-ARMA model with truncated assumption
I-ACD-ARMA	The ACD-ARMA model with independent assumption
T-DEN-ARMA	The DENSITY-ARMA model with truncated assumption
I-DEN-ARMA	The DENSITY-ARMA model with independent assumption
T-ACD-DEN	The ACD-DENSITY model with truncated assumption
I-ACD-DEN	The ACD-DENSITY model with independent assumption

Table 7.3: The performance of the proposed order placement strategies at several risk aversion levels when used to make order placement decisions for the IBM stock traded in the New York Stock Exchange. The performance reported includes the probability that the submitted order is executed (P_E), the average profit obtained when the submitted order is executed ($E(U_E)$), the average loss incurred when the submitted order is not executed ($E(U_{NE})$), the average profit obtained ($E(U)$), and the variance of the profit ($V(U)$).

λ	Buy						Sell					
	P_E	$E(U_E)$	$E(U_{NE})$	$E(\Delta)$	$E(U)$	$V(U)$	P_E	$E(U_E)$	$E(U_{NE})$	$E(\Delta)$	$E(U)$	$V(U)$
Truncated ACD-ARMA Model												
0.0	0.49	0.0499	-0.0406	0.37	0.0035	0.0381	0.38	0.1604	-0.0927	0.15	0.0041	0.0401
0.5	0.86	0.0230	-0.1196	0.05	0.0025	0.0118	0.41	0.1485	-0.0963	0.14	0.0047	0.0373
1.0	0.93	0.0135	-0.1635	0.02	0.0019	0.0062	0.44	0.1387	-0.0987	0.13	0.0047	0.0350
1.5	0.96	0.0083	-0.1863	0.01	0.0013	0.0034	0.46	0.1303	-0.1007	0.12	0.0045	0.0334
2.0	0.98	0.0054	-0.1945	0.01	0.0009	0.0022	0.47	0.1223	-0.1022	0.11	0.0042	0.0312
2.5	0.99	0.0035	-0.2069	0.00	0.0005	0.0015	0.49	0.1151	-0.1031	0.11	0.0044	0.0294
3.0	0.99	0.0023	-0.2111	0.00	0.0001	0.0011	0.51	0.1078	-0.1030	0.10	0.0043	0.0273
3.5	0.99	0.0016	-0.2227	0.00	0.0000	0.0008	0.52	0.1011	-0.1026	0.10	0.0043	0.0255
4.0	1.00	0.0011	-0.2507	0.00	-0.0001	0.0007	0.54	0.0948	-0.1023	0.09	0.0039	0.0240
Independent ACD-ARMA Model												
0.0	0.42	0.1484	-0.0988	0.15	0.0048	0.0389	0.41	0.1505	-0.0959	0.15	0.0044	0.0393
0.5	0.54	0.1045	-0.1107	0.11	0.0046	0.0288	0.53	0.1051	-0.1081	0.10	0.0043	0.0291
1.0	0.64	0.0748	-0.1188	0.08	0.0052	0.0210	0.63	0.0744	-0.1163	0.08	0.0044	0.0215
1.5	0.72	0.0550	-0.1257	0.06	0.0053	0.0157	0.72	0.0542	-0.1234	0.06	0.0041	0.0158
2.0	0.79	0.0412	-0.1327	0.04	0.0046	0.0121	0.79	0.0402	-0.1277	0.04	0.0046	0.0117
2.5	0.84	0.0312	-0.1391	0.03	0.0045	0.0090	0.84	0.0302	-0.1345	0.03	0.0044	0.0091
3.0	0.88	0.0237	-0.1427	0.03	0.0043	0.0064	0.88	0.0228	-0.1419	0.02	0.0038	0.0071
3.5	0.92	0.0180	-0.1478	0.02	0.0039	0.0048	0.92	0.0172	-0.1455	0.02	0.0034	0.0054
4.0	0.94	0.0136	-0.1499	0.01	0.0032	0.0036	0.94	0.0129	-0.1522	0.01	0.0029	0.0036
Truncated Density-ARMA Model												
0.0	0.04	0.1918	-0.0075	0.62	0.0006	0.0508	0.39	0.1450	-0.0860	0.14	0.0036	0.0322
0.5	0.41	0.0132	-0.0081	0.21	0.0006	0.0113	0.42	0.1342	-0.0892	0.13	0.0036	0.0307
1.0	0.67	0.0034	-0.0075	0.09	-0.0002	0.0036	0.44	0.1258	-0.0909	0.12	0.0038	0.0293
1.5	0.88	0.0008	-0.0055	0.03	0.0000	0.0010	0.46	0.1178	-0.0924	0.11	0.0040	0.0277
2.0	0.97	0.0002	-0.0042	0.00	0.0001	0.0002	0.48	0.1104	-0.0923	0.11	0.0042	0.0256
2.5	1.00	0.0000	-0.0315	0.00	-0.0001	0.0001	0.50	0.1021	-0.0922	0.10	0.0043	0.0232
3.0	1.00	0.0000	0.3400	0.00	0.0000	0.0000	0.51	0.0954	-0.0913	0.10	0.0041	0.0215
3.5	1.00	0.0000	0.3400	0.00	0.0000	0.0000	0.52	0.0896	-0.0892	0.09	0.0044	0.0192
4.0	1.00	0.0000	0.3400	0.00	0.0000	0.0000	0.54	0.0833	-0.0883	0.09	0.0045	0.0176
Independent Density-ARMA Model												
0.0	0.35	0.1450	-0.0758	0.15	0.0023	0.0291	0.35	0.1460	-0.0743	0.15	0.0031	0.0293
0.5	0.47	0.0883	-0.0732	0.10	0.0027	0.0178	0.46	0.0905	-0.0713	0.11	0.0036	0.0177
1.0	0.56	0.0612	-0.0714	0.08	0.0032	0.0121	0.55	0.0630	-0.0699	0.08	0.0035	0.0126
1.5	0.63	0.0448	-0.0683	0.06	0.0034	0.0087	0.63	0.0460	-0.0678	0.06	0.0039	0.0091
2.0	0.70	0.0333	-0.0644	0.04	0.0035	0.0059	0.70	0.0346	-0.0654	0.05	0.0042	0.0065
2.5	0.75	0.0249	-0.0614	0.03	0.0032	0.0042	0.75	0.0262	-0.0616	0.03	0.0043	0.0046
3.0	0.80	0.0187	-0.0582	0.03	0.0031	0.0031	0.80	0.0199	-0.0565	0.03	0.0043	0.0032
3.5	0.84	0.0137	-0.0537	0.02	0.0029	0.0021	0.83	0.0147	-0.0517	0.02	0.0037	0.0023
4.0	0.88	0.0096	-0.0472	0.01	0.0026	0.0014	0.87	0.0104	-0.0445	0.01	0.0032	0.0016
Truncated ACD-Density Model												
0.0	0.71	0.0703	-0.1569	0.10	0.0041	0.0350	0.37	0.1633	-0.0897	0.15	0.0041	0.0392
0.5	0.81	0.0502	-0.2023	0.07	0.0032	0.0280	0.42	0.1446	-0.0954	0.13	0.0045	0.0358
1.0	0.88	0.0361	-0.2489	0.05	0.0021	0.0219	0.45	0.1296	-0.1000	0.12	0.0042	0.0331
1.5	0.92	0.0261	-0.2892	0.03	0.0007	0.0175	0.49	0.1175	-0.1042	0.11	0.0044	0.0308
2.0	0.94	0.0192	-0.3252	0.02	-0.0001	0.0142	0.52	0.1074	-0.1088	0.10	0.0042	0.0293
2.5	0.96	0.0145	-0.3615	0.02	-0.0002	0.0116	0.55	0.0985	-0.1123	0.09	0.0042	0.0278
3.0	0.97	0.0111	-0.3848	0.01	-0.0003	0.0095	0.61	0.0871	-0.1227	0.09	0.0045	0.0262
3.5	0.98	0.0086	-0.4198	0.01	-0.0002	0.0078	0.66	0.0766	-0.1360	0.08	0.0049	0.0249
4.0	0.98	0.0068	-0.4614	0.01	-0.0006	0.0068	0.71	0.0670	-0.1489	0.07	0.0047	0.0234
Independent ACD-Density Model												
0.0	0.43	0.1457	-0.1007	0.14	0.0048	0.0387	0.42	0.1473	-0.0974	0.14	0.0046	0.0391
0.5	0.61	0.0967	-0.1348	0.10	0.0068	0.0322	0.60	0.0972	-0.1314	0.10	0.0052	0.0328
1.0	0.75	0.0671	-0.1711	0.07	0.0064	0.0262	0.74	0.0666	-0.1668	0.07	0.0050	0.0268
1.5	0.84	0.0473	-0.2102	0.05	0.0050	0.0212	0.83	0.0463	-0.2045	0.05	0.0039	0.0217
2.0	0.89	0.0340	-0.2443	0.04	0.0026	0.0178	0.89	0.0328	-0.2435	0.04	0.0021	0.0181
2.5	0.92	0.0249	-0.2749	0.03	0.0015	0.0146	0.93	0.0237	-0.2787	0.03	0.0010	0.0150
3.0	0.94	0.0186	-0.3037	0.02	0.0004	0.0122	0.95	0.0176	-0.3158	0.02	0.0005	0.0123
3.5	0.96	0.0141	-0.3289	0.02	-0.0004	0.0103	0.96	0.0133	-0.3484	0.02	-0.0004	0.0106
4.0	0.97	0.0109	-0.3582	0.01	-0.0006	0.0087	0.97	0.0103	-0.3749	0.01	-0.0004	0.0088

Table 7.4: The performance of the proposed order placement strategies and the best static strategy when used to make order placement decisions for the instruments in the Multi Commodity Exchange of India. The performance reported includes the probability that the submitted order is executed (P_E), the average profit obtained when the submitted order is executed ($E(U_E)$), the average loss incurred when the submitted order is not executed ($E(U_{NE})$), the average profit obtained ($E(U)$), and the gain/loss in percentage when compared to the best static strategy.

Strategy	Δ/λ	P_E	Training dataset				Testing dataset				
			$E(U_E)$	$E(U_{NE})$	$E(U)$	%	P_E	$E(U_E)$	$E(U_{NE})$	$E(U)$	%
Liquidity buy problem for Gold future, MCX											
MARKET	-	1.00	-2.6356	-	-2.6356	-	1.00	-3.3537	-	-3.3537	-
LMARKET	-	0.00	-	-2.7558	-2.7558	-	0.00	-	-3.593	-3.593	-
BEST LIMIT	7	0.55	4.0956	-9.9886	-2.263	-	0.60	3.2913	-12.5233	-2.9682	-
T-ACD-ARMA	0.002	0.58	0.6447	-6.0006	-2.1432	5.30%	0.56	1.2604	-8.286	-2.8942	2.50%
I-ACD-ARMA	0.038	0.73	1.5695	-12.8164	-2.2903	-1.20%	0.73	1.3562	-14.8049	-2.928	1.40%
T-DEN-ARMA	0.007	0.48	-0.8093	-3.5061	-2.2011	2.70%	0.54	-1.5367	-4.9033	-3.0948	-4.30%
I-DEN-ARMA	0.035	0.67	1.2568	-9.5154	-2.2726	-0.40%	0.7	0.3866	-11.5807	-3.1913	-7.50%
T-ACD-DEN	0.000	0.62	3.1335	-11.2481	-2.3453	-3.60%	0.59	4.4427	-13.6369	-2.8997	2.30%
I-ACD-DEN	0.009	0.58	4.3987	-11.5322	-2.2203	1.90%	0.6	4.6666	-14.122	-2.9291	1.30%
Liquidity sell problem for Gold future, MCX											
MARKET	-	1.00	-2.6356	-	-2.6356	-	1.00	-3.3537	-	-3.3537	-
LMARKET	-	0.00	-	-2.5233	-2.5233	-	0.00	-	-3.0558	-3.0558	-
BEST LIMIT	5	0.68	2.1814	-10.937	-2.0635	-	0.72	1.3887	-13.0614	-2.5963	-
T-ACD-ARMA	0.049	0.55	4.9488	-10.5892	-2.0948	-1.50%	0.59	4.577	-12.5053	-2.505	3.50%
I-ACD-ARMA	0.026	0.61	3.6299	-11.1175	-2.1133	-2.40%	0.62	3.4711	-13.2599	-2.917	-12.40%
T-DEN-ARMA	0.058	0.57	3.9071	-10.0436	-2.0708	-0.40%	0.64	3.0458	-12.1469	-2.4361	6.20%
I-DEN-ARMA	0.014	0.45	5.535	-8.3021	-2.055	0.40%	0.51	4.6932	-10.2455	-2.6442	-1.80%
T-ACD-DEN	0.057	0.63	3.6295	-11.6495	-2.071	-0.40%	0.65	3.369	-13.8379	-2.5926	0.10%
I-ACD-DEN	0.006	0.51	5.9787	-10.5162	-2.0823	-0.90%	0.52	6.4013	-12.6331	-2.6534	-2.20%
Liquidity buy problem for Silver future, MCX											
MARKET	-	1.00	-13.9272	-	-13.9272	-	1.00	-8.5941	-	-8.5941	-
LMARKET	-	0.00	-	-14.2661	-14.2661	-	0.00	-	-8.7152	-8.7152	-
BEST LIMIT	20	0.4	2.7885	-22.6995	-12.4615	-	0.48	10.3945	-23.2267	-6.9312	-
T-ACD-ARMA	0.000	0.43	-4.3234	-17.9896	-12.1312	2.70%	0.39	3.5043	-14.3259	-7.3623	-6.20%
I-ACD-ARMA	0.018	0.48	2.8293	-25.2498	-11.8178	5.20%	0.64	3.2918	-24.9985	-6.8902	0.60%
T-DEN-ARMA	0.001	0.23	-1.6784	-15.6905	-12.4859	-0.20%	0.28	3.4333	-11.7572	-7.4675	-7.70%
I-DEN-ARMA	0.014	0.4	4.2479	-22.8506	-12.0092	3.60%	0.59	3.8189	-22.7046	-7.1371	-3.00%
T-ACD-DEN	0.002	0.62	-3.1283	-28.0195	-12.6417	-1.40%	0.55	6.8026	-24.8378	-7.5635	-9.10%
I-ACD-DEN	0.000	0.37	6.7799	-23.5323	-12.3283	1.10%	0.40	16.3573	-22.8207	-7.1582	-3.30%
Liquidity sell problem for Silver future, MCX											
MARKET	-	1.00	-13.9272	-	-13.9272	-	1.00	-8.5941	-	-8.5941	-
LMARKET	-	0.00	-	-13.6492	-13.6492	-	0.00	-	-8.2202	-8.2202	-
BEST LIMIT	27	0.31	8.9971	-20.8138	-11.4747	-	0.37	16.9567	-21.2201	-7.2121	-
T-ACD-ARMA	0.000	0.41	6.8507	-23.8936	-11.3018	1.50%	0.43	14.6263	-23.783	-7.1659	0.60%
I-ACD-ARMA	0.014	0.43	6.6679	-24.0288	-10.8055	5.80%	0.59	5.452	-23.8278	-6.5993	8.50%
T-DEN-ARMA	0.000	0.41	4.4635	-22.6947	-11.5549	-0.70%	0.46	12.3014	-23.0055	-6.7662	6.20%
I-DEN-ARMA	0.012	0.38	6.1091	-21.9234	-11.1369	2.90%	0.57	4.8188	-21.1451	-6.4428	10.70%
T-ACD-DEN	0.000	0.38	7.8089	-23.1627	-11.3751	0.90%	0.43	15.01	-23.4589	-7.0858	1.80%
I-ACD-DEN	0.000	0.37	8.873	-22.9993	-11.3011	1.50%	0.4	16.2643	-22.7785	-7.1268	1.20%
Liquidity buy problem for Natural Gas future, MCX											
MARKET	-	1.00	-0.9031	-	-0.9031	-	1.00	-0.3633	-	-0.3633	-
LMARKET	-	0.00	-	-0.8942	-0.8942	-	0.00	-	-0.3607	-0.3607	-
BEST LIMIT	0.9	0.23	-0.3671	-1.0106	-0.8611	-	0.12	0.2589	-0.43	-0.349	-
T-ACD-ARMA	0.033	0.43	-0.7164	-0.8718	-0.8058	6.40%	0.40	-0.2267	-0.3785	-0.3176	9.00%
I-ACD-ARMA	0.085	0.12	0.5026	-1.0248	-0.8415	2.30%	0.13	0.3754	-0.4558	-0.3468	0.60%
T-DEN-ARMA	0.089	0.14	-0.5275	-0.8815	-0.8335	3.20%	0.07	0.0857	-0.3715	-0.3374	3.30%
I-DEN-ARMA	0.084	0.11	0.5007	-1.0135	-0.845	1.90%	0.09	0.4614	-0.4348	-0.3539	-1.40%
T-ACD-DEN	0.037	0.56	-0.555	-1.2675	-0.8677	-0.80%	0.69	-0.2281	-0.6399	-0.3541	-1.40%
I-ACD-DEN	0.000	0.32	-0.3219	-1.1052	-0.8573	0.40%	0.27	0.0627	-0.4867	-0.3366	3.60%
Liquidity sell problem for Natural Gas future, MCX											
MARKET	-	1.00	-0.9031	-	-0.9031	-	1.00	-0.3633	-	-0.3633	-
LMARKET	-	0.00	-	-0.9105	-0.9105	-	0.00	-	-0.3681	-0.3681	-
BEST LIMIT	1.000	0.23	-0.2868	-0.989	-0.8261	-	0.81	-0.2775	-0.6328	-0.3454	-
T-ACD-ARMA	0.017	0.43	-0.3247	-1.1818	-0.8091	2.10%	0.44	0.027	-0.5473	-0.2965	14.20%
I-ACD-ARMA	0.085	0.14	0.576	-1.0346	-0.8105	1.90%	0.17	0.4076	-0.458	-0.3141	9.10%
T-DEN-ARMA	0.000	0.49	-0.5099	-1.1232	-0.8252	0.10%	0.36	-0.0069	-0.4731	-0.3044	11.90%
I-DEN-ARMA	0.085	0.13	0.5403	-1.0176	-0.8174	1.10%	0.09	0.5206	-0.4232	-0.3383	2.10%
T-ACD-DEN	0.010	0.37	-0.2891	-1.1268	-0.8138	1.50%	0.32	0.0991	-0.4959	-0.3074	11.00%
I-ACD-DEN	0.000	0.37	-0.2485	-1.1365	-0.8109	1.80%	0.35	0.077	-0.5132	-0.3048	11.80%

Table 7.5: The performance of the proposed order placement strategies and the best static strategy when apply to make order placement decision for the instruments in the New York Stock Exchange. The performance reported include the probability that the submitted order is executed (P_E), the average profit obtained when the submitted order is executed ($E(U_E)$), the average loss incurred when the submitted order is not executed ($E(U_{NE})$), the average profit obtained ($E(U)$), and the gain/loss in percentage when comparing to the best static strategy.

Strategy	Δ/λ	P_E	Training dataset					Testing dataset			
			$E(U_E)$	$E(U_{NE})$	$E(U)$	%	P_E	$E(U_E)$	$E(U_{NE})$	$E(U)$	%
Liquidity buy problem for GE stock, NYSE											
MARKET	-	1.00	0.0000	-	0.0000	-	1.00	0.0000	-	0.0000	-
LMARKET	-	0.00	-	0.0002	0.0002	-	0.00	-	0.0005	0.0005	-
BEST LIMIT	0.02	0.80	0.0200	-0.0354	0.0092	-	0.87	0.0200	-0.0683	0.0085	-
T-ACD-ARMA	0.20	0.78	0.0186	-0.0290	0.0081	-11.89%	0.73	0.0329	-0.0620	0.0068	-19.63%
I-ACD-ARMA	5.20	0.80	0.0211	-0.0409	0.0090	-1.66%	0.81	0.0276	-0.0781	0.0073	-14.73%
T-DEN-ARMA	3.00	0.91	0.0057	-0.0088	0.0043	-52.56%	0.96	0.0018	-0.0124	0.0013	-84.77%
I-DEN-ARMA	5.00	0.81	0.0172	-0.0294	0.0082	-10.16%	0.91	0.0110	-0.0449	0.0057	-32.88%
T-ACD-DEN	0.00	0.70	0.0263	-0.0334	0.0084	-8.55%	0.60	0.0538	-0.0646	0.0063	-25.70%
I-ACD-DEN	2.40	0.75	0.0259	-0.0407	0.0091	-0.77%	0.70	0.0436	-0.0762	0.0071	-16.38%
Liquidity sell problem for GE stock, NYSE											
MARKET	-	1.00	0.0000	-	0.0000	-	1.00	0.0000	-	0.0000	-
LMARKET	-	0.00	-	-0.0002	-0.0002	-	0.00	-	-0.0005	-0.0005	-
BEST LIMIT	0.02	0.80	0.0200	-0.0365	0.0090	-	0.86	0.0200	-0.0773	0.0067	-
T-ACD-ARMA	11.00	0.59	0.0346	-0.0321	0.0073	-18.50%	0.65	0.0435	-0.0638	0.0063	-5.30%
I-ACD-ARMA	4.40	0.79	0.0224	-0.0416	0.0087	-2.76%	0.78	0.0299	-0.0765	0.0063	-5.93%
T-DEN-ARMA	8.60	0.61	0.0322	-0.0308	0.0075	-16.23%	0.71	0.0339	-0.0642	0.0057	-14.29%
I-DEN-ARMA	4.80	0.82	0.0168	-0.0305	0.0084	-6.00%	0.91	0.0107	-0.0461	0.0057	-15.19%
T-ACD-DEN	11.00	0.74	0.0258	-0.0399	0.0085	-4.74%	0.72	0.0380	-0.0768	0.0062	-6.82%
I-ACD-DEN	2.60	0.77	0.0240	-0.0431	0.0085	-5.20%	0.70	0.0419	-0.0784	0.0060	-9.68%
Liquidity buy problem for IBM stock, NYSE											
MARKET	-	1.00	0.0000	-	0.0000	-	1.00	0.0000	-	0.0000	-
LMARKET	-	0.00	-	-0.0004	-0.0004	-	0.00	-	-0.0029	-0.0029	-
BEST LIMIT	0.08	0.57	0.0800	-0.0975	0.0041	-	0.60	0.0800	-0.1159	0.0019	-
T-ACD-ARMA	0.00	0.49	0.0499	-0.0406	0.0035	-16.58%	0.49	0.0422	-0.0374	0.0015	-18.71%
I-ACD-ARMA	1.50	0.72	0.0550	-0.1257	0.0053	27.62%	0.72	0.0566	-0.1377	0.0020	6.98%
T-DEN-ARMA	0.20	0.20	0.0365	-0.0074	0.0014	-67.44%	0.16	0.0435	-0.0103	-0.0014	-175.92%
I-DEN-ARMA	1.20	0.59	0.0538	-0.0696	0.0037	-9.63%	0.63	0.0599	-0.1008	0.0011	-41.32%
T-ACD-DEN	0.00	0.71	0.0703	-0.1569	0.0041	-0.68%	0.67	0.0664	-0.1303	0.0015	-19.47%
I-ACD-DEN	0.60	0.64	0.0899	-0.1414	0.0068	63.15%	0.62	0.0860	-0.1337	0.0034	76.13%
Liquidity sell problem for IBM stock, NYSE											
MARKET	-	1.00	0.0000	-	0.0000	-	1.00	0.0000	-	0.0000	-
LMARKET	-	0.00	-	0.0004	0.0004	-	0.00	-	0.0029	0.0029	-
BEST LIMIT	0.07	0.62	0.0700	-0.1015	0.0043	-	0.66	0.0700	-0.1189	0.0049	-
T-ACD-ARMA	0.80	0.43	0.1426	-0.0978	0.0049	11.67%	0.45	0.1314	-0.0964	0.0062	25.98%
I-ACD-ARMA	0.20	0.46	0.1309	-0.1013	0.0048	10.36%	0.47	0.1234	-0.0989	0.0059	20.35%
T-DEN-ARMA	5.20	0.58	0.0708	-0.0850	0.0048	9.77%	0.61	0.0787	-0.1070	0.0057	17.38%
I-DEN-ARMA	2.80	0.78	0.0222	-0.0585	0.0044	2.13%	0.84	0.0207	-0.0879	0.0036	-27.36%
T-ACD-DEN	3.50	0.66	0.0766	-0.1360	0.0049	12.26%	0.67	0.0760	-0.1345	0.0057	17.35%
I-ACD-DEN	0.40	0.56	0.1052	-0.1238	0.0053	21.91%	0.56	0.1008	-0.1172	0.0058	17.88%
Liquidity buy problem for Microsoft stock, NYSE											
MARKET	-	1.00	0.0000	-	0.0000	-	1.00	0.0000	-	0.0000	-
LMARKET	-	0.00	-	0.0007	0.0007	-	0.00	-	-0.0005	-0.0005	-
BEST LIMIT	0.02	0.82	0.0200	-0.0415	0.0092	-	0.83	0.0200	-0.0427	0.0091	-
T-ACD-ARMA	0.00	0.77	0.0225	-0.0401	0.0081	-12.30%	0.79	0.0201	-0.0371	0.0083	-9.07%
I-ACD-ARMA	6.20	0.81	0.0218	-0.0470	0.0089	-3.29%	0.82	0.0207	-0.0444	0.0088	-2.98%
T-DEN-ARMA	2.60	0.94	0.0063	-0.0188	0.0047	-48.88%	0.94	0.0065	-0.0221	0.0048	-47.46%
I-DEN-ARMA	5.00	0.84	0.0159	-0.0316	0.0081	-12.23%	0.84	0.0167	-0.0355	0.0082	-9.80%
T-ACD-DEN	0.80	0.75	0.0268	-0.0496	0.0079	-13.89%	0.77	0.0242	-0.0455	0.0079	-12.61%
I-ACD-DEN	3.00	0.74	0.0286	-0.0475	0.0087	-5.90%	0.74	0.0264	-0.0442	0.0083	-8.46%
Liquidity sell problem for Microsoft stock, NYSE											
MARKET	-	1.00	0.0000	-	0.0000	-	1.00	0.0000	-	0.0000	-
LMARKET	-	0.00	-	-0.0007	-0.0007	-	0.00	-	0.0005	0.0005	-
BEST LIMIT	0.02	0.83	0.0200	-0.0444	0.0088	-	0.84	0.0200	-0.0429	0.0098	-
T-ACD-ARMA	11.00	0.67	0.0235	-0.0314	0.0055	-37.31%	0.68	0.0232	-0.0299	0.0061	-37.13%
I-ACD-ARMA	1.80	0.75	0.0184	-0.0331	0.0057	-36.03%	0.75	0.0177	-0.0259	0.0069	-29.02%
T-DEN-ARMA	10.20	0.56	0.0294	-0.0255	0.0053	-40.35%	0.56	0.0313	-0.0261	0.0063	-35.74%
I-DEN-ARMA	1.00	0.68	0.0153	-0.0205	0.0037	-58.08%	0.68	0.0150	-0.0162	0.0049	-49.97%
T-ACD-DEN	14.80	0.78	0.0249	-0.0515	0.0084	-4.73%	0.79	0.0239	-0.0458	0.0094	-3.31%
I-ACD-DEN	3.40	0.75	0.0273	-0.0500	0.0081	-8.30%	0.77	0.0257	-0.0447	0.0092	-5.84%

Table 7.6: The profit gained from using the proposed strategy over immediately executing the trade at the beginning of the trading period in number of ticks averaged over all instruments in the same markets together with the improvement over the best static strategy in percentage terms and its ranking.

Model	MCX Training			MCX Testing			NYSE Training			NYSE Testing		
	$E(U)$	%	#	$E(U)$	%	#	$E(U)$	%	#	$E(U)$	%	#
BEST LIMIT	1.0088	0.00%	6	0.7516	0.00%	6	0.7445	0.00%	2	0.6811	0.00%	1
T-ACD-ARMA	1.2281	21.74%	2	0.8491	12.96%	2	0.6218	-16.47%	5	0.5884	-13.61%	5
I-ACD-ARMA	1.2735	26.25%	1	0.8698	15.72%	1	0.7063	-5.13%	3	0.6207	-8.88%	3
T-DEN-ARMA	1.0481	3.90%	5	0.8301	10.43%	3	0.4660	-37.41%	7	0.3732	-45.22%	7
I-DEN-ARMA	1.1818	17.15%	3	0.8042	6.99%	5	0.6106	-17.99%	6	0.4859	-28.67%	6
T-ACD-DEN	0.9899	-1.86%	7	0.7343	-2.31%	7	0.7044	-5.38%	4	0.6205	-8.90%	4
I-ACD-DEN	1.0956	8.61%	4	0.8135	8.23%	4	0.7738	3.93%	1	0.6635	-2.59%	2

improvement over the best static strategy in training and testing dataset, respectively. For the NYSE market, the ACD-DENSITY model with the independent assumption is the best performing model with 3.93% improvement in the training dataset. The result also indicates that the model with independent assumption generally performed better than the model with the truncated assumption (except in the case of the DENSITY-ARMA model in the MCX testing dataset).

Introducing market variables

We now add market variables into our model to investigate whether adding market variables can improve the order placement decision or not. To achieve this, we modify the ACD and the ARMA-GARCH model in Equation (7.24) and (7.28) to include the effect from market variables by adding the market variables into the conditional mean equations resulting in

$$\begin{aligned}
 \psi_i^{\delta_1} = & \omega + \sum_{j=1}^p \alpha_j \psi_{i-j}^{\delta_1} (|\epsilon_{i-j} - b| + c_j (\epsilon_{i-j} - b))^{\delta_2} \\
 & + \sum_{j=1}^p \lambda_j (|\epsilon_{i-j} - b| + c_j (\epsilon_{i-j} - b))^{\delta_2} + \sum_{j=1}^q \beta_j \psi_{i-j}^{\delta_1} + \sum_{j=1}^s \gamma_j x_{ij}, \quad (7.38)
 \end{aligned}$$

for the conditional mean function of the ACD model, and

$$\hat{r}_i = \omega + \sum_{j=1}^p \alpha_j r_{i-j} + \sum_{j=1}^q \beta_j z_{i-j} + z_i + \sum_{j=1}^s \gamma_j x_{ij}, \quad (7.39)$$

for the conditional mean function of the ARMA-GARCH model when $x_i = (x_{i1}, \dots, x_{is})$ is the market variables at the i -th time step.

Since we only have market variables for the MCX dataset, this section will perform the analysis on only the MCX dataset. The market variables considered here consist of the bid-ask spread, order imbalance (the difference between the number of orders at the best bid and the best ask), and trading volume in the previous trading period, which is reported to improve trading performance in Nevmyvaka et al. [72]. Table 7.7 displays the performance of the best risk aversion level for each of the six models together with the three static strategies mentioned above. Similar to the case of no market variables, the result indicates that all models beat the static strategies that always execute the trade at the beginning and at the end of the trading period in all situations. Additionally, the best of these six models also beat

Table 7.7: The performance of the proposed order placement strategies with market variables and the best static strategy when used to make order placement decision for the instruments in the Multi Commodity Exchange of India. The performance reported includes the probability that the submitted order is executed (P_E), the average profit obtained when the submitted order is executed ($E(U_E)$), the average loss incurred when the submitted order is not executed ($E(U_{NE})$), the average profit obtained ($E(U)$), and the gain/loss in percentage when compared to the best static strategy.

Strategy	Δ/λ	P_E	Training dataset				P_E	$E(U_E)$	Testing dataset		
			$E(U_E)$	$E(U_{NE})$	$E(U)$	%			$E(U_{NE})$	$E(U)$	%
Liquidity buy problem for Gold future, MCX											
MARKET	-	1.00	-2.6356	-	-2.6356	-	1.00	-3.3537	-	-3.3537	-
LMARKET	-	0.00	-	-2.7558	-2.7558	-	0.00	-	-3.5930	-3.5930	-
BEST LIMIT	7	0.55	4.0956	-9.9886	-2.2630	-	0.60	3.2913	-12.5233	-2.9682	-
T-ACD-ARMA	0.008	0.73	0.1681	-8.7285	-2.2025	2.68%	0.70	0.3004	-10.7919	-3.0363	-2.30%
I-ACD-ARMA	0.031	0.67	2.5107	-11.9494	-2.2839	-0.92%	0.69	1.9915	-14.3339	-3.0576	-3.01%
T-DEN-ARMA	0.015	0.76	-1.2868	-4.8906	-2.1687	4.17%	0.75	-1.9302	-6.2305	-2.9847	-0.56%
I-DEN-ARMA	0.028	0.61	1.9138	-9.0821	-2.3455	-3.64%	0.67	0.7044	-11.0116	-3.2038	-7.94%
T-ACD-DEN	0.016	0.83	0.4393	-15.7548	-2.3717	-4.80%	0.81	0.6914	-18.8440	-3.1136	-4.90%
I-ACD-DEN	0.03	0.84	0.5771	-16.5422	-2.2369	1.16%	0.50	6.8490	-12.8132	-2.9387	0.99%
Liquidity sell problem for Gold future, MCX											
MARKET	-	1.00	-2.6356	-	-2.6356	-	1.00	-3.3537	-	-3.3537	-
LMARKET	-	0.00	-	-2.5233	-2.5233	-	0.00	-	-3.0558	-3.0558	-
BEST LIMIT	5	0.68	2.1814	-10.9370	-2.0635	-	0.72	1.3887	-13.0614	-2.5963	-
T-ACD-ARMA	0.064	0.58	4.4270	-10.9370	-2.0640	-0.02%	0.62	4.0155	-13.0351	-2.5072	3.43%
I-ACD-ARMA	0.021	0.57	4.4582	-10.5350	-2.0437	0.96%	0.57	4.5130	-12.5243	-2.8389	-9.35%
T-DEN-ARMA	0.052	0.55	4.1547	-9.7784	-2.0518	0.57%	0.62	3.3729	-12.1278	-2.4542	5.47%
I-DEN-ARMA	0.016	0.49	4.4640	-8.3040	-2.0455	0.88%	0.54	3.6905	-10.2512	-2.6644	-2.62%
T-ACD-DEN	0.068	0.66	3.2422	-12.2083	-2.0573	0.30%	0.69	2.9486	-14.5518	-2.5219	2.86%
I-ACD-DEN	0.009	0.57	5.1129	-11.3937	-2.0596	0.19%	0.58	5.4707	-13.6478	-2.6033	-0.27%
Liquidity buy problem for Silver future, MCX											
MARKET	-	1.00	-13.9272	-	-13.9272	-	1.00	-8.5941	-	-8.5941	-
LMARKET	-	0.00	-	-14.2661	-14.2661	-	0.00	-	-8.7152	-8.7152	-
BEST LIMIT	20	0.40	2.7885	-22.6995	-12.4615	-	0.48	10.3945	-23.2267	-6.9312	-
T-ACD-ARMA	0.005	0.62	-5.3800	-21.5245	-11.4349	8.24%	0.72	-2.5691	-18.1243	-6.9101	0.30%
I-ACD-ARMA	0.013	0.41	5.4125	-24.3563	-12.0391	3.39%	0.60	4.9270	-25.3493	-7.1992	-3.87%
T-DEN-ARMA	0.009	0.55	-6.4325	-18.2981	-11.7399	5.79%	0.69	-4.2680	-14.9368	-7.5683	-9.19%
I-DEN-ARMA	0.007	0.32	6.3043	-21.6261	-12.6626	-1.61%	0.48	7.8295	-22.2245	-7.7134	-11.29%
T-ACD-DEN	0.003	0.65	-2.5233	-29.4288	-11.8876	4.61%	0.67	3.4547	-28.7167	-7.2335	-4.36%
I-ACD-DEN	0.001	0.39	7.8801	-24.5098	-11.8030	5.28%	0.48	13.5446	-25.3791	-6.8440	1.26%
Liquidity sell problem for Silver future, MCX											
MARKET	-	1.00	-13.9272	-	-13.9272	-	1.00	-8.5941	-	-8.5941	-
LMARKET	-	0.00	-	-13.6492	-13.6492	-	0.00	-	-8.2202	-8.2202	-
BEST LIMIT	27	0.31	8.9971	-20.8138	-11.4747	-	0.37	16.9567	-21.2201	-7.2121	-
T-ACD-ARMA	0	0.39	9.8776	-24.1011	-10.9930	4.20%	0.43	15.4360	-24.2853	-7.1006	1.55%
I-ACD-ARMA	0.013	0.42	7.9824	-24.1931	-10.8167	5.73%	0.59	5.9884	-24.3773	-6.5772	8.80%
T-DEN-ARMA	0.000	0.40	6.2994	-23.2220	-11.3821	0.81%	0.46	12.8073	-23.0657	-6.6454	7.86%
I-DEN-ARMA	0.008	0.32	11.1002	-21.6620	-11.0348	3.83%	0.50	8.8558	-20.9699	-6.1176	15.18%
T-ACD-DEN	0.003	0.41	7.8032	-24.0072	-11.1120	3.16%	0.47	13.1287	-24.8965	-7.0700	1.97%
I-ACD-DEN	0.003	0.43	6.8151	-24.9905	-11.2268	2.16%	0.49	12.0850	-25.7921	-7.2800	-0.94%
Liquidity buy problem for Natural Gas future, MCX											
MARKET	-	1.00	-0.9031	-	-0.9031	-	1.00	-0.3633	-	-0.3633	-
LMARKET	-	0.00	-	-0.8942	-0.8942	-	0.00	-	-0.3607	-0.3607	-
BEST LIMIT	0.9	0.23	-0.3671	-1.0106	-0.8611	-	0.12	0.2589	-0.4300	-0.3490	-
T-ACD-ARMA	0.060	0.47	-0.5156	-1.0321	-0.7883	8.46%	0.44	-0.1711	-0.4289	-0.3155	9.61%
I-ACD-ARMA	0.089	0.14	0.4654	-1.0485	-0.8292	3.72%	0.15	0.3827	-0.4574	-0.3355	3.89%
T-DEN-ARMA	0.089	0.11	-0.2569	-0.9115	-0.8391	2.56%	0.07	0.1522	-0.3761	-0.3382	3.10%
I-DEN-ARMA	0.060	0.10	0.5543	-1.0061	-0.8451	1.87%	0.07	0.6389	-0.4248	-0.3540	-1.43%
T-ACD-DEN	0.013	0.59	-0.3882	-1.4561	-0.8217	4.58%	0.69	-0.1953	-0.6540	-0.3360	3.74%
I-ACD-DEN	0.000	0.34	-0.1294	-1.1801	-0.8231	4.41%	0.28	0.1146	-0.5031	-0.3280	6.02%
Liquidity sell problem for Natural Gas future, MCX											
MARKET	-	1.00	-0.9031	-	-0.9031	-	1.00	-0.363332	-	-0.363332	-
LMARKET	-	0.00	-	-0.9105	-0.9105	-	0.00	-	-0.36812	-0.36812	-
BEST LIMIT	1	0.23	-0.2868	-0.9890	-0.8261	-	0.81	-0.277526	-0.63279	-0.345426	-
T-ACD-ARMA	0.003	0.41	-0.1576	-1.2298	-0.7938	3.92%	0.41	0.0691839	-0.552794	-0.294706	14.68%
I-ACD-ARMA	0.088	0.14	0.5725	-1.0417	-0.8127	1.63%	0.16	0.403118	-0.451998	-0.313196	9.33%
T-DEN-ARMA	0.037	0.48	-0.4544	-1.1477	-0.8166	1.16%	0.39	0.0041958	-0.49375	-0.299728	13.23%
I-DEN-ARMA	0.074	0.12	0.6390	-1.0152	-0.8181	0.98%	0.10	0.490945	-0.425961	-0.335306	2.93%
T-ACD-DEN	0.012	0.36	-0.1664	-1.1599	-0.8028	2.82%	0.28	0.14509	-0.487378	-0.309381	10.43%
I-ACD-DEN	0.009	0.34	-0.1339	-1.1535	-0.8038	2.71%	0.29	0.135054	-0.49147	-0.309537	10.39%

Table 7.8: Comparison between the performance of the best strategy with market variables and the one without market variables.

Instrument	Side	Training dataset			Testing dataset		
		$E(U)$ with X	$E(U)$ no X	Gain(%)	$E(U)$ with X	$E(U)$ no X	Gain(%)
Gold futures	Buy	-2.1687	-2.1432	-1.19%	-2.9847	-2.8942	-3.13%
Gold futures	Sell	-2.0437	-2.0550	0.55%	-2.8389	-2.6442	-7.36%
Silver futures	Buy	-11.4349	-11.8178	3.24%	-6.9101	-6.8902	-0.29%
Silver futures	Sell	-10.8167	-10.8055	-0.10%	-6.5772	-6.5993	0.33%
Natural gas futures	Buy	-0.7883	-0.8058	2.17%	-0.3155	-0.3176	0.66%
Natural gas futures	Sell	-0.7938	-0.8091	1.89%	-0.2947	-0.2965	0.61%
Average ticks gained		-7.0475	-7.1618	1.60%	-4.2355	-4.1948	-0.97%

the best static strategy in all situations as well.

To compare the improvement gained from including market variables into the model, Table 7.8 compares the performance of the best performing models that includes market variables and the one that does not include market variables. The result indicates that including market variables into the model does not necessarily improve trading performance as we gain performance improvement in the training dataset in only four out of the six cases considered. Additionally, we gain improvement both in the training dataset and testing dataset only in two cases for the Natural gas futures. When considering the average profit gained from including the market variables into the model, averaged over all situations, we find that the model with market variables has better performance in the training dataset while it performs worse in the testing dataset.

Consequently, the result studied in this section indicates that including market variables in our model does not always result in better trading performance, as suggested in [72], and it is more appropriate to try the model with and without market variables to select the best model for making the order placement decision. A general guideline to select the best model for a particular instrument is to adopt the model that has the best combined performance on the buy and the sell problem. By using this criterion, we will select the model without market variables for Gold futures, while we will select the model with market variables for Silver and Natural gas futures which is exactly the best performing model for each instrument.

7.5 Summary

This chapter proposed a new framework for making order placement decisions based on the trade-off between the profit gained from better execution price and the risk of non-execution in a mean-variance optimisation framework. This framework is general enough to solve all trading problems mentioned in Harris [42] as traders can define their trading objective by specifying two payoff functions: a function $U_E(p)$ that defines the payoff that traders will get when they execute the trade at price level p , and a function $U_{NE}(p)$ that defines the cost that traders need to pay when their order is not executed and the price of the asset at the end of the trading period is p . In particular, the order placement problem of liquidity traders who need to transact their order before some specified deadline can be modelled by setting $U_{NE}(p) = U_E(p)$ so that the cost that a trader needs to pay if his order is not executed is equal to the cost of executing the trade at the end of the trading period. The utility function of informed traders

and value motivated traders can be modelled by setting $U_{NE}(p) = \max\{U_E(p), U_{NT}\}$, where U_{NT} is a utility gained from not trading, so that they will trade at the end of the period only when it is profitable to do so. After specifying the objective function, traders also need to specify a model for the probability that the limit order at price level p^L will be executed before the end of the period and a probability density function of the asset price at the end of the trading period given no execution of limit order at price level p^L . Accordingly, we discuss three different approaches to model these two probabilities, which are i) the unconditional model implied by an arithmetic Brownian motion, ii) the empirical unconditional model using density estimation and iii) the empirical conditional distribution using ACD and ARMA-GARCH models.

To measure the performance of the proposed framework in making order placement decisions, we performed an experiment to apply the proposed framework to make order placement decision for liquidity traders using the historical data from the Multi Commodity Exchange of India (MCX) and the New York Stock Exchange (NYSE). The results indicated that the proposed framework beats the simple static strategies that always execute the trade at the beginning of the period and at the end of the period in all cases studied. The proposed framework also beat the best static strategy in all cases in the MCX dataset, while it beat the best static strategy only in two out of six cases considered in the NYSE dataset. Although the proposed framework could not beat the best static strategy in all cases, the improvement gained from the proposed framework when it can beat the best static strategy is very significant. Additionally, the result obtained from the model with market variables indicated that adding market variables into the model does not necessary improve the trading performance of the model. This suggests that it is more appropriate to try both models and select the one that performs best in the training dataset to make the trading decision.